national center for ontological research ncor

University at Buffalo The State University of New York

Ontology Foundry Efforts

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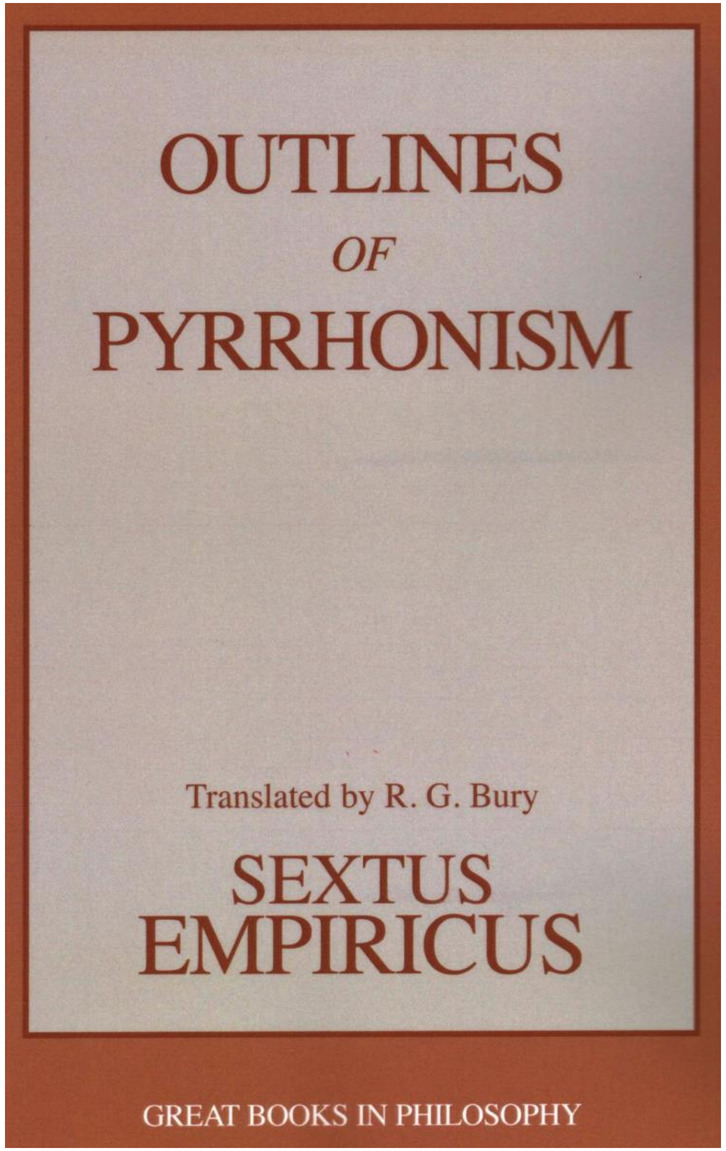
Pyrrhonism

- An ancient skeptical philosophy that taught suspension of belief was the only way to maintain an inquisitive attitude

- They were known for giving arguments on either side of claim, in the interest of showing there was no real truth

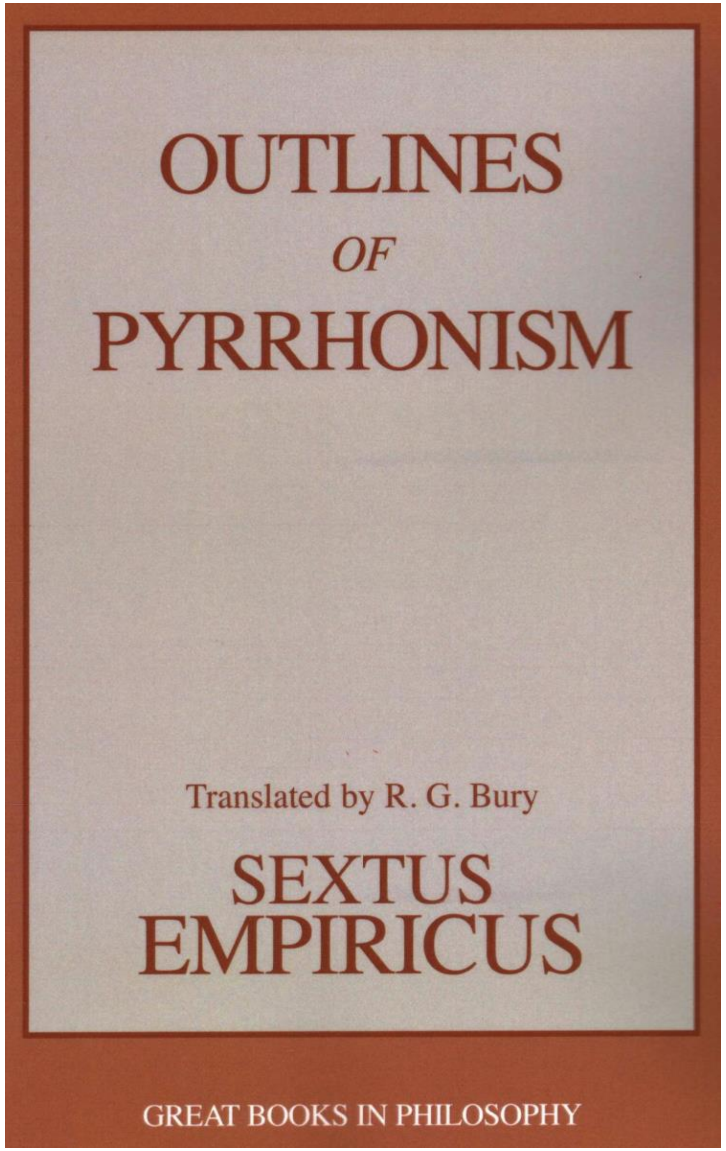
- They were also known for giving **bad arguments** on occassion



Pyrrhonism

- In his defense of Pyrronism, Sextus Empiricus wrote on this point:

- Physicians do not always give the strongest medicine, for the patient’s body is not ready; they instead give weaker doses, building a patient’s constitution, so they are **ready for the strongest medicine**



Building Knowledge Models Can be Easy

Building a Knowledge Graph From Scratch Using LLMs

Turn your Pandas data frame into a knowledge graph using LLMs. Build your own LLM graph-builder from scratch, implement LLMGraphTransformer by LangChain, and QA your KG.

Cristian Leo Follow 36 min read · Nov 25, 2024

How to Build a Knowledge Graph in Minutes (And Make It Enterprise-Ready)

I tried and failed creating one—but it was when LLMs were not a thing!

Thuwarakesh Murallie Follow 8 min read · Jan 13, 2025

Constructing knowledge graphs from text using OpenAI functions

Seamlessy implement information extraction pipeline with LangChain and Neo4j

Tomaz Bratanic Follow 11 min read · Oct 20, 2023

Building Knowledge Models Can be Easy

AWS Database Blog

Build a knowledge graph in Amazon Neptune using Data Lens

by Russell Waterson | on 05 MAY 2021 | in Amazon Neptune | Permalink | Comments | Share

This is a guest post by Russell Waterson, Knowledge Graph Engineer at Data Lens Ltd.

Customers use knowledge graphs to consolidate and integrate information assets a

Building knowledge graphs by getting data from disparate existing data sources car
complex. Project planning, project management, engineering, maintenance and rele
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The Data Lens Ltd. team have been implementing Knowledge Graph solutions as a
now released tooling to greatly reduce the time and effort required to build a Know

Use Data Lens to make building a knowledge graph in Amazon Neptune faster and
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How to build a knowledge graph in 7 steps

John Stegeman, Kara Doriani O'Shee

March 12, 2025 14 min read

SILOS

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Building ontologies or knowledge graphs according to standards can be HARD

But that's how we avoid silos…

Disciplined Semantics to Governance

- Ontology methodology clarifies what kinds of entities and relations we are talking about

- BFO supplies top-level discipline but once many communities build together, the challenge becomes coordination, validation, and reuse

- And identifying **infrastructure that lets ontologies function as a community asset**

Guiding Questions

How much alignment can you enforce across communities without undermining user goals?

What can be done to minimize downstream alignment efforts across diverging communities?

Guiding Questions

How much alignment can you enforce across communities without undermining user goals?

What can be done to minimize downstream alignment efforts across diverging communities?

It Takes a Village

- The same label can designate different kinds of things in the same community, e.g. **solitude**, **target**, **biomarker**

- The same format can carry different assumptions about identity, time, evidence, and context

- Data can be exported while the meaning needed to interpret it remains inside a platform, workflow, or codebase

Smithery

- Coordinate **meaning**: identifiers, definitions, relations, and axioms
- Coordinate **method**: modeling principles, reuse expectations, and quality criteria
- Coordinate **lifecycle**: review, validation, release, deprecation, and maintenance
- Coordinate **community**: domain experts, ontology engineers, users, and governance

Open Biological and Biomedical Ontologies

- In 2005, a consortium of biologists decided to create standards for ontology development

- Such as requiring ontologies be open-source, have documentation, include definitions for vocabulary terms...

Overview
Open (principle 1)
Common Format (principle 2)
URI/Identifier Space (principle 3)
Versioning (principle 4)
Scope (principle 5)
Textual Definitions (principle 6)
Relations (principle 7)
Documentation (principle 8)
Documented Plurality of Users (principle 9)
Commitment To Collaboration (principle 10)
Locus of Authority (principle 11)
Naming Conventions (principle 12)
Notification of Changes (principle 13)
Maintenance (principle 16)
Responsiveness (principle 20)

Industrial Ontologies Foundry

- In 2016, the Systems Integration Division at the National Institute of Standards and Technology (NIST) brought together industrial manufacturing researchers and ontologists.

- Who aimed to promote best practices for ontology development and preserve interoperability.

1 Introduction
2 IOF Ontology Architecture
3 Common Representations
4 Open Availability and Copyright
5 Scope and Context
6 Modularity
7 Ontological Considerations
8 Foundational Ontology – Use
9 Documentation
10 Textual Definitions
11 Identifiers and Naming Conventions
12 IRI and Identifier Space
13 Versioning
14 Maintenance
15 Contributed Ontologies: – Use
16 Collaboration

National Security Ontology Foundry

- 2016 also witnessed chartered efforts to promote interoperability among ontologies created within and across USG; formally recognized in 2023 by IC and DOD chief data officers
- Aimed at constructing vetted ontology artifacts aligned with best practices, tailored to specific USG missions

Federation

National Security Ontology ~~Foundry~~

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Ontology “Foundries”

Open Biological and Biomedical Foundry

Industrial Ontologies Foundry

National Security Ontology Federation

FY26 National Defense Authorization Act

SEC. 1504. DEPARTMENT OF DEFENSE DATA ONTOLOGY GOVERNANCE WORKING GROUP.

(a) Establishment.—

(1) In general.—The Secretary of Defense shall establish a working group to develop and implement a common data ontology and governance structure across the Department of Defense.

(2) Designation.—The working group established under to paragraph (1) shall be known as the "Department of Defense Data Ontology Governance Working Group" (in this section the "Working Group").

(3) Use of existing structures.—

(A) In general.—Notwithstanding paragraph (1), the Secretary of Defense may designate an existing forum, council, or organizational body to serve as the Working Group if such entity satisfies the requirements of subsections (b) and (c).

(B) Rule of construction.—For the purposes of this section, a forum, council, or organizational body designated under subparagraph (A) is deemed to be a working group established by the Secretary of Defense under paragraph (1).

(b) Purpose.—The purpose of the Working Group is to inform and to progress the Department of Defense's foundational data ontology work by developing and implementing domain-specific data ontologies and governance structures across the Department of Defense to expand data interoperability, enhance information

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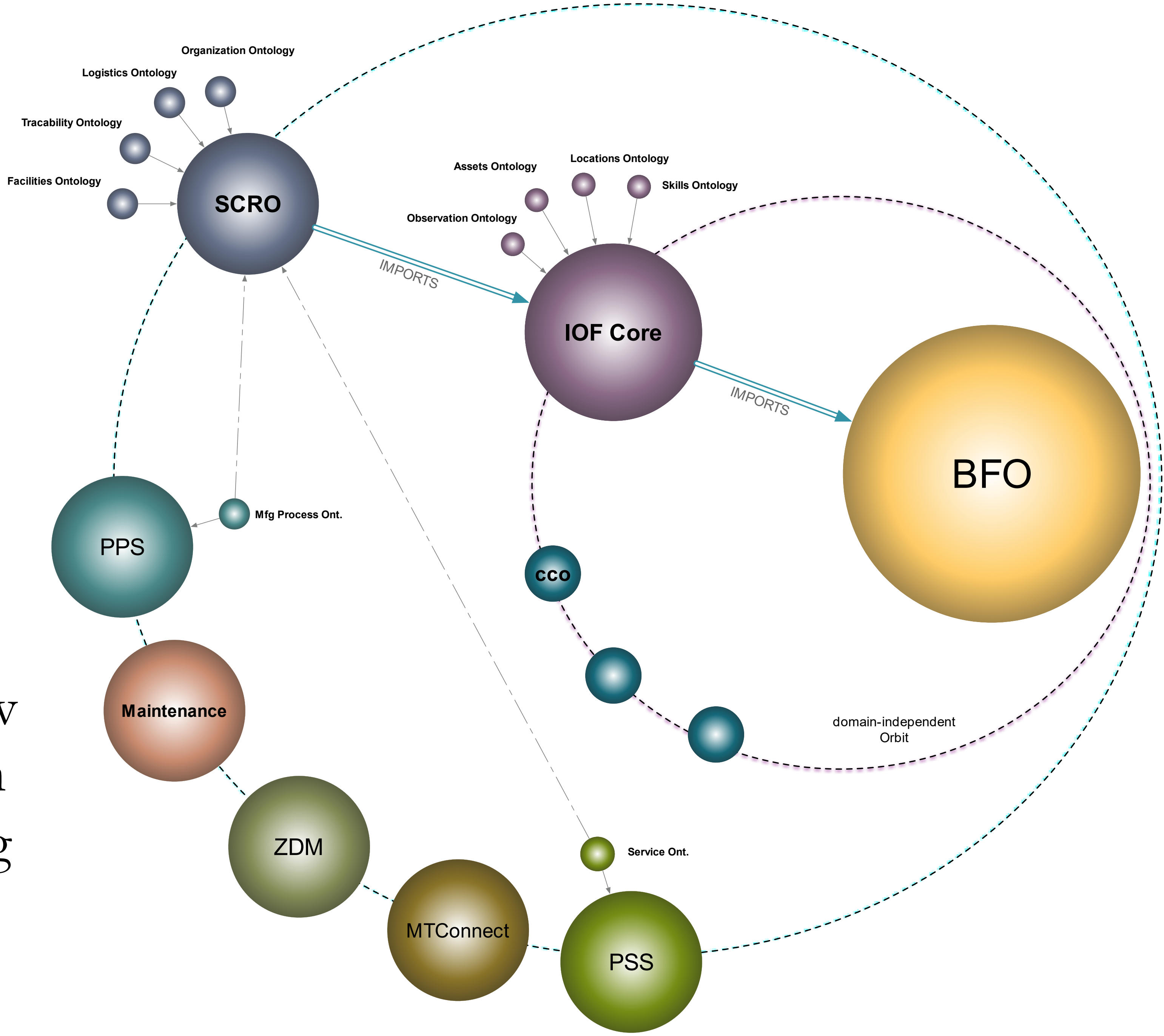
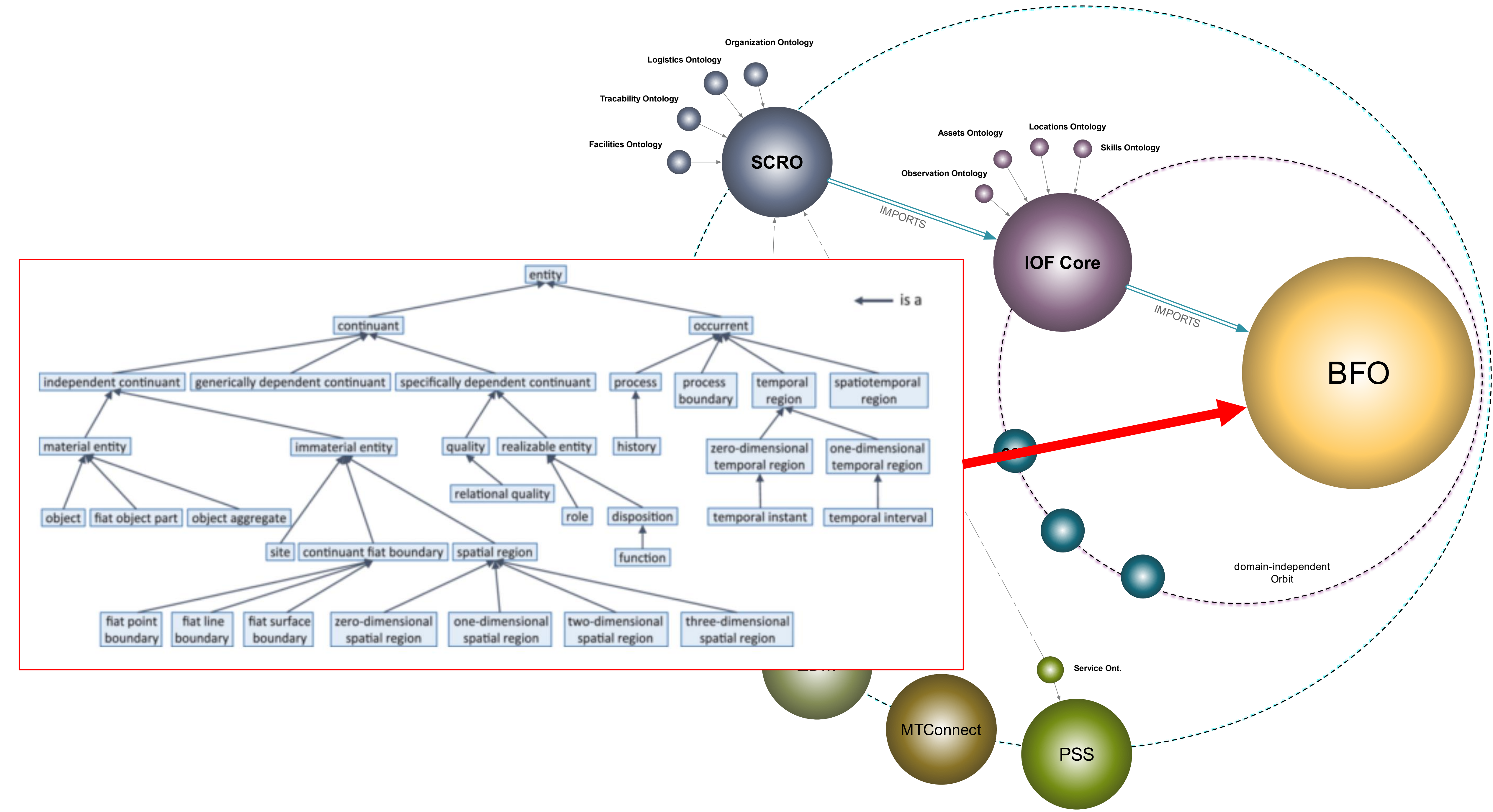
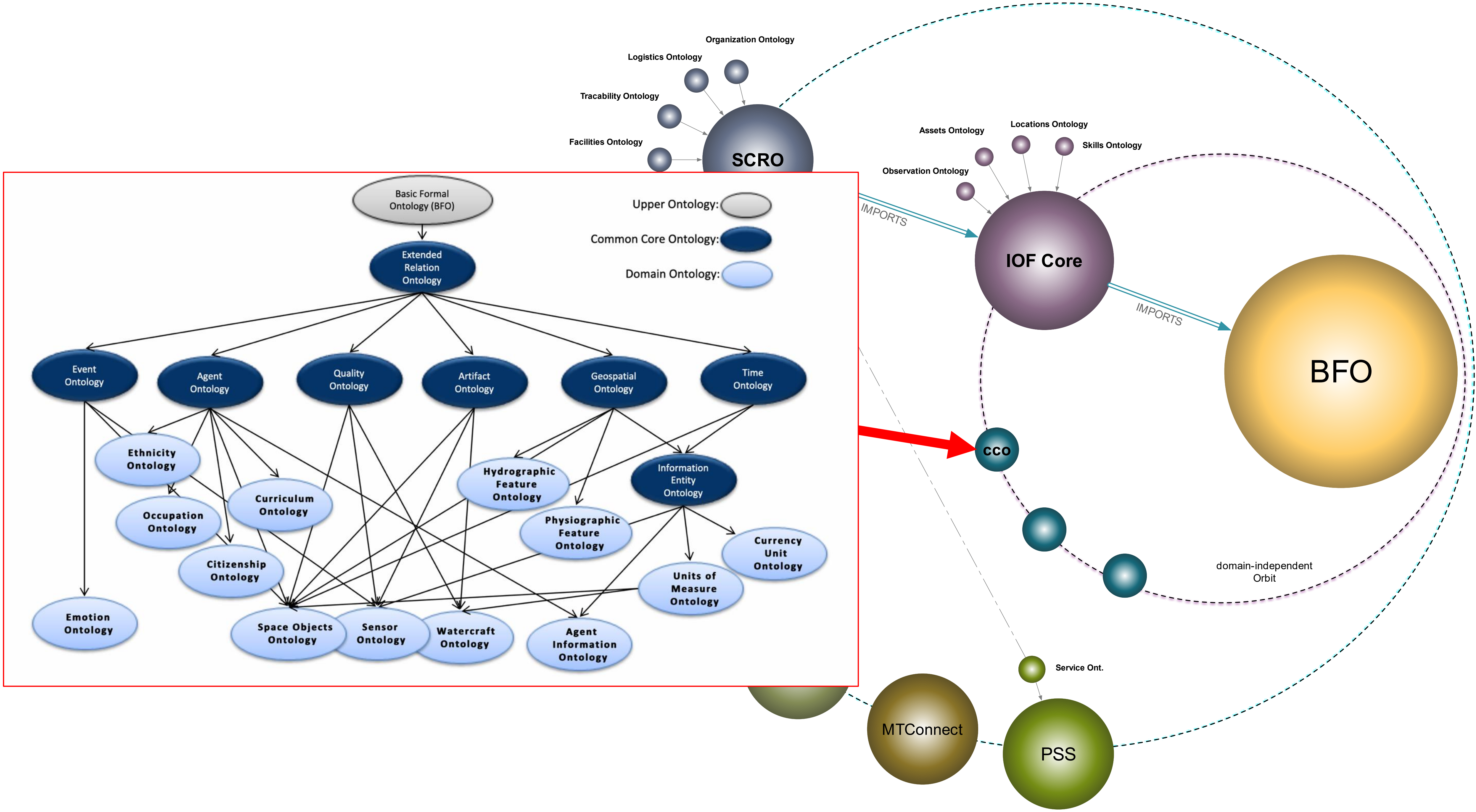
Enforcement

Ontologies in these Foundries are hubs logically extending from higher-level ontologies

Ontologies are extended by **downward population**, new classes have parent classes in a hierarchy ultimately leading to a BFO class




Organization Ontology
Logistics Ontology
Tracability Ontology
Facilities Ontology
SCRO
IMPORTS
Assets Ontology
Locations Ontology
Skills Ontology
Observation Ontology
IOF Core
IMPORTS
BFO
domain-independent
Orbit
Service Ont.
MTConnect
PSS
is a
entity
continuant
occurrent
independent continuant
generically dependent continuant
specifically dependent continuant
process
process boundary
temporal region
spatiotemporal region
material entity
immaterial entity
quality
realizable entity
history
zero-dimensional temporal region
one-dimensional temporal region
relational quality
object
fiat object part
object aggregate
role
disposition
temporal instant
temporal interval
site
continuant fiat boundary
spatial region
function
fiat point boundary
fiat line boundary
fiat surface boundary
zero-dimensional spatial region
one-dimensional spatial region
two-dimensional spatial region
three-dimensional spatial region



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cco
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Service Ont.
MTConnect
PSS
Basic Formal Ontology (BFO)
Upper Ontology:
Common Core Ontology:
Domain Ontology:
Extended Relation Ontology
Event Ontology
Agent Ontology
Quality Ontology
Artifact Ontology
Geospatial Ontology
Time Ontology
Ethnicity Ontology
Curriculum Ontology
Occupation Ontology
Citizenship Ontology
Hydrographic Feature Ontology
Information Entity Ontology
Physiographic Feature Ontology
Currency Unit Ontology
Units of Measure Ontology
Emotion Ontology
Space Objects Ontology
Sensor Ontology
Watercraft Ontology
Agent Information Ontology

BFO Ecosystem

800+ Projects

BFO Basic Formal Ontology

Home GitHub Guidebook Publications FOL Users Tutorials

Users

Below you will find an alphabetical list of ontologies and institutions/groups usin

Ontologies

- ACGT Master Ontology (ACGT MO)
- Actionable Intelligence Retrieval System (AIRS)
- Addiction Ontology (Addict-O)
- Additive Manufacturing Ontology (AMO)
- Adolescent Depression Ontology (ADO)
- Adverse Event Reporting Ontology (AERO)
- AFO Foundational Ontology
- African Wildlife Ontology (AWO)
- Agronomic Linked Data (AgroLD)
- Agronomy Ontology (AGRO)
- Aircraft System Ontology
- Algorithm-Implementation-Execution Ontology Design Pattern
- Alzheimer Disease Ontology (ADO)
- Alzheimer's Disease Diagnosis Ontology (ADDO)
- Anatomy of the Insect Skeleto-Muscular system ontology (AISM)
- Animals in Context Ontology (ACO)
- Anthropological Notation Ontology (ANNO)
- Antimicrobial-Microorganism Ontology
- Apollo Structured Vocabulary (Apollo-SV)
- Argument Ontology (ARGO)
- ARIES (Arkansas Imaging Enterprise System) Knowledge Graph
- Asset Management Ontology (AMODO)
- Autism-DSM-ADI-R Ontology (ADAR)
- Bacterial Clinical Infectious Diseases Ontology (BCIDO)
- Baden Württemberg Materials Digital Domain Ontology (BWMD)
- Bank Ontology
- Battle Management Ontology (BMO)
- Behavior Change Intervention Ontology (BCIO)
- Behaviour Change Technique Ontology
- Behavior Perspective Model (BPM)
- Beta Cell Genomics Application Ontology (BCGO)
- Bio-Knowledge Network Ontology (BioKNO)
- BioAssay Ontology
- Bioinformatics Web Service Ontology (OBIWS)
- Biological Collections Ontology (BCO)
- Biomedical Ethics Ontology
- Biomedical Grid Terminology (BiomedGT, retired)
- Biomedical Study - Lifecycle Management (BMS-LM) core ontology
- Biomimetic Ontology
- BioTop: a biomedical top-domain ontology

- OntoAlign++: A Combined Strategy for Improving Ontologies Alignment
- OntoBuildableSpace Ontology
- OntoDM Core
- OntoForInfoScience
- Ontologies for Representing Surgical Procedure Models (OntoSPM)
- Ontologized Minimum Information About Biobank data Sharing (OMIAB
- Ontology Based Clinical Decision Support System for Geriatrics
- Ontology Based Decision Support System for Tuberculosis Manageme
- Ontology for Adverse Events (OAE)
- Ontology for Autism Spectrum Disorder
- Ontology for Biobanking (OBIB)
- Ontology for Biofilms (BIFO)
- Ontology for Biomedical Investigations (OBI)
- Ontology for Cancer research variables (OCRV)
- Ontology for Computable Eligibility Criteria - Hepatitis C Virus (OCEC-H
- Ontology for Dengue Fever (IDODEN)
- Ontology for Documentation of Variable/Data Source Selection (ODVD
- Ontology for Drug Discovery Investigations (DDI)
- Ontology for Energy Investigations (OEI)
- Ontology for Functionally Graded Materials (OFGM)
- Ontology for General Medical Science (OGMS)
- Ontology for Genes and Genomes - Mouse (OGG-MM)
- Ontology for Genetic Interval (OGI)
- Ontology for Guiding Appropriate Antibiotic Prescribing
- Ontology for Information Science (OntoforInforScience)
- Ontology for Laparoscopic Surgeries (LapOntoSPM)
- Ontology for MIcroRNA Target Prediction (OMIT) (here)
- Ontology for Newborn Screening and Translational Research (ONSTR
- Ontology for Next Generation Sequencing Experiments (NGS Ontology
- Ontology for Nutritional Epidemiology (ONE)
- Ontology for Nutritional Studies (ONS)
- Ontology for Pain and Related Disability, Mental Health and Quality of
- Ontology for Parasite LifeCycle (OPL)
- Ontology for Periodontitis (PERIO)
- Ontology for Petroleum Production
- Ontology for Prognostic Health Management (PHM) in Spacecraft Avio
- Ontology for Stem Cell Investigations (OSCI)
- Ontology for the Documentation of Variable Selection and Data source
- Ontology for Thoracentesis
- Ontology of Autonomous Driving Based on the SAE J3016 Standard
- Ontology of Arthropod Circulatory Systems (OArCS)
- Ontology of Biological and Clinical Statistics (OBCS)
- Ontology of Cancer Related Social-Ecological Variables (OCRSEV)
- Ontology of Card Sleights
- Ontology of Cardiovascular Drug Adverse Events (OCVDAE)
- Ontology of Chinese Medicine for Rheumatism (OCMR)
- Ontology of Clinical Research (OCRe)
- Ontology of Commercial Exchange (OCE)
- Ontology of Data Mining (OntoDM)

- Shop-Floor Digital Twin (DT) ontology
- Situated and Interactive Multimodal Conversations
- Situation Awareness Ontology (SAO)
- Sketch Map Ontology
- Skin Physiology Ontology (SPO)
- Sleep Domain Ontology (SDO)
- SMART Protocols: SeMAntic RepresenTation for Experimental Protocols
- Smart Ultrasound in Obstetrics and Gynecology (SUOG) Ontology
- SNOMED CT (SCT) Standard Ontology
- Social Determinants of Health Ontology (SDoHO)
- Social Psychology Ontology (SPO)
- Socialog Ontology for Social Simulation
- Software Ontology (SWO)
- Software, Disabilities and Competences Ontology (SODIC)
- Soil Food Web Ontology
- Space Domain Ontologies (SDO)
- Space Object Ontology (SOO)
- Spatial Graph Adapter (SGA) Ontology Design Pattern
- Spatial Relation Ontology
- Spatiotemporal Ontology for the Administrative Units of Switzerland (SONADUS
- Special Nuclear Materials Detection Ontology (SNM-DO)
- Statistics Ontology (STATO)
- STATO-LMM Linear Mixed Model Ontology
- Style of Delivery Ontology
- Subcellular Anatomy Ontology (SAO)
- Suggested Ontology for Pharmacogenomics (SO-Pharm)
- Supply Chain Traceability Ontology
- Surface Water Ontology (SWO)
- Survey Ontology
- Sustainable Development and Climate (SDC) Ontology
- Symptomatic Treatment of Multiple Sclerosis Ontology (STMSO)
- Taxonomy for Rehabilitation of Knee Conditions (TRAK)
- The Common Rule Ontology (CRO)
- The Trope Ontology
- Time Event Ontology (TEO)
- Toxic Process Ontology (TXPO)
- Trade-Space Analysis Tool for Constellations (TAT-C) ontology
- Traditional Chinese Drugs Ontology (TCDO)
- Translational Medicine Ontology (TMO)

https://bfo-ontology.github.io/users.html

Guiding Questions

How much alignment can you enforce across communities without undermining user goals?

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Policy, Funding, Inspired Volunteers, Technological Artifacts

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Guiding Questions

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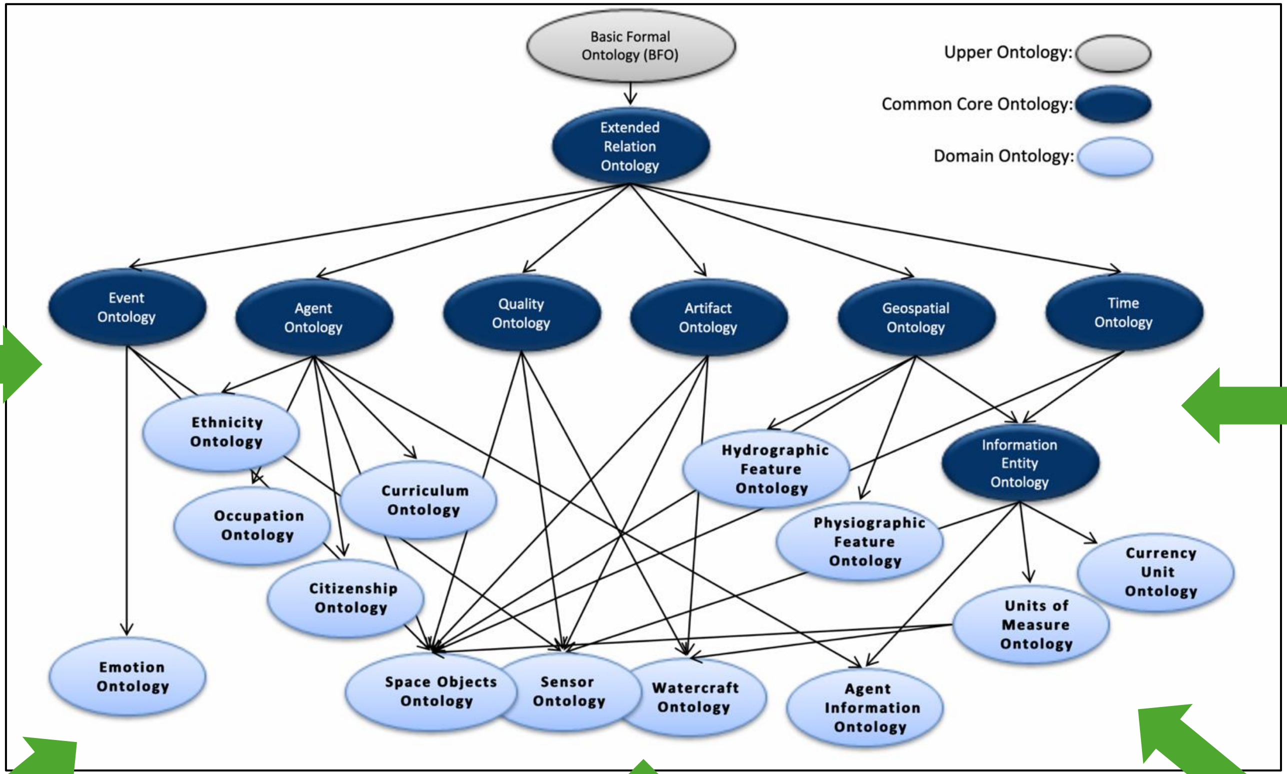
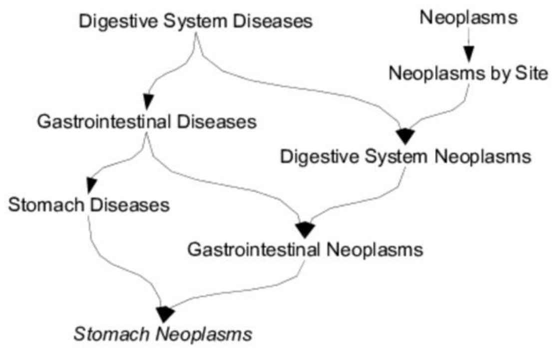
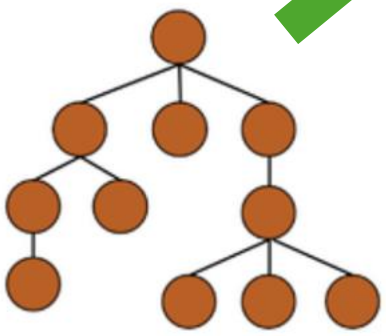
Bag of Words

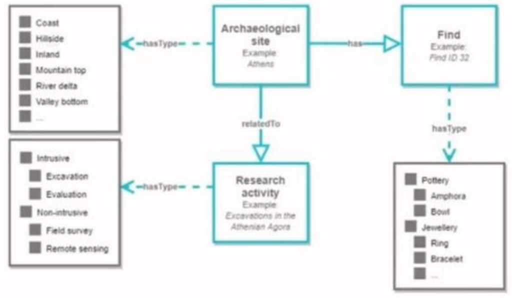




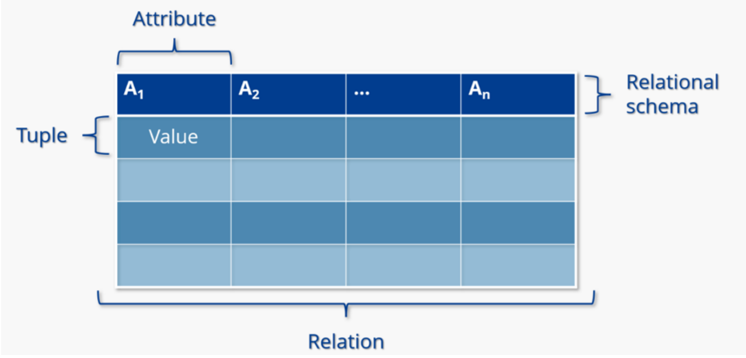
Thesaurus



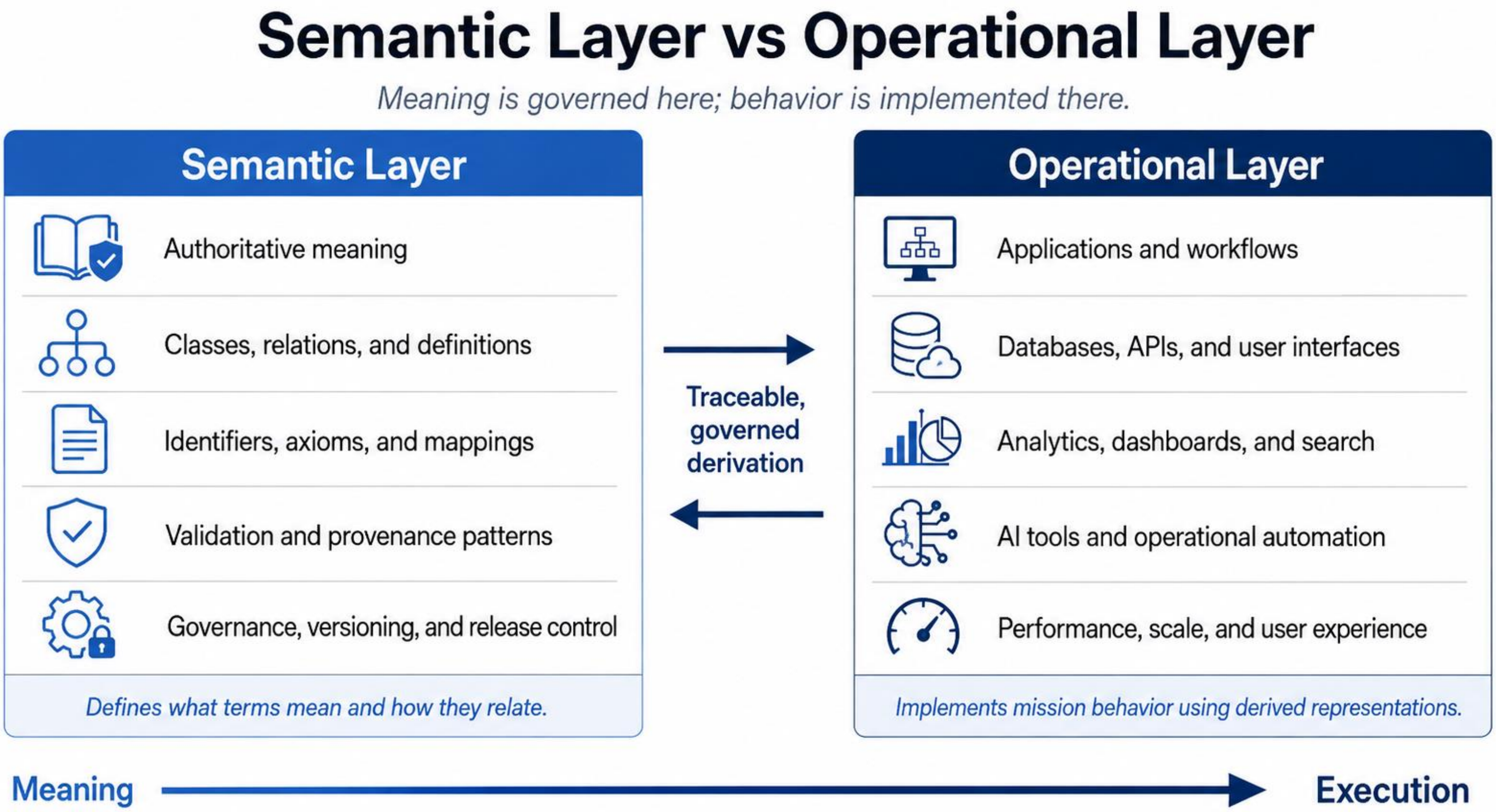
Ontology



Controlled Vocabulary



Relational Database


Semantic Layer vs Operational Layer
Meaning is governed here; behavior is implemented there.
Semantic Layer
Authoritative meaning
Classes, relations, and definitions
Identifiers, axioms, and mappings
Validation and provenance patterns
Governance, versioning, and release control
Defines what terms mean and how they relate.
Traceable, governed derivation
Operational Layer
Applications and workflows
Databases, APIs, and user interfaces
Analytics, dashboards, and search
AI tools and operational automation
Performance, scale, and user experience
Implements mission behavior using derived representations.
Meaning
Execution

Semantic Layer

- An authoritative semantic layer facilitates traceable derived products
- Derived products should remain accountable to governed meaning
- Failure mode allows convenience artifacts to be ungoverned authorities

Authoritative semantic layer

classes · relations
definitions · axioms

→

Derived semantic products

overlays · mappings
profiles · AI artifacts

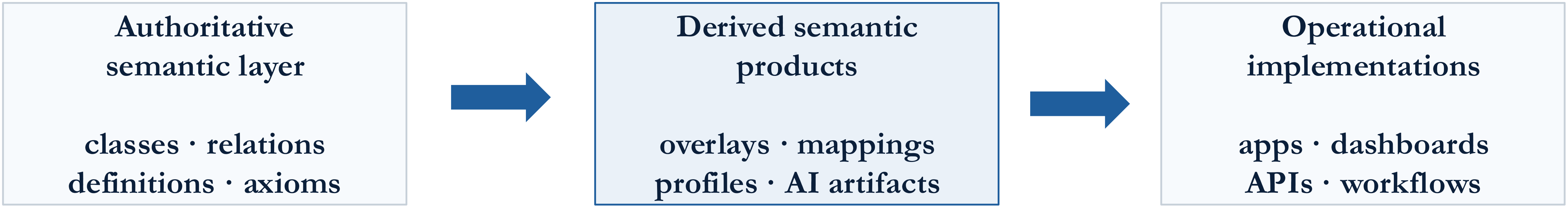
→

Operational implementations

apps · dashboards
APIs · workflows

Derived Products

- Operational systems can use whatever runtime representations they need
- If they depend on governed meaning, they must remain traceable to it
- The governing question is whether meaning survives derivation



Derived Products

- Applications often need categories for search, navigation, workflow routing, etc.

- Some categories **may not** be in the authoritative semantic at all

- A governed relation can simulate hierarchy-like behavior so application gets the hierarchy it needs and the reference ontology remains clean

Derived Products

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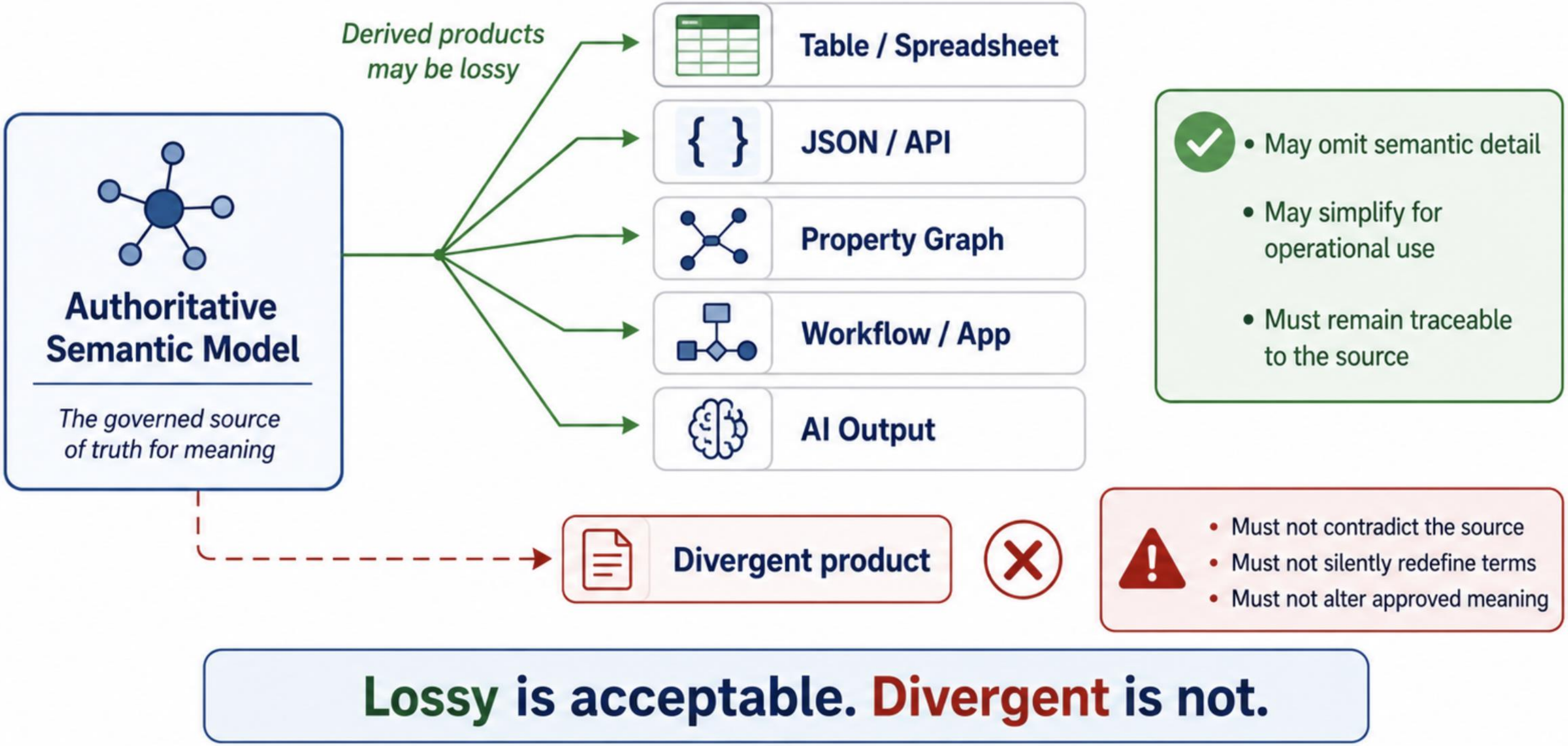
Spoiler: John is going to teach you the logic you need to do this later


Derived products
may be lossy
Authoritative
Semantic Model
The governed source
of truth for meaning
Table / Spreadsheet
JSON / API
Property Graph
Workflow / App
AI Output
May omit semantic detail
May simplify for operational use
Must remain traceable to the source
Divergent product
Must not contradict the source
Must not silently redefine terms
Must not alter approved meaning
Lossy is acceptable. Divergent is not.

Summary

- We need governance around ontology development and maintenance to make good on the promises of ontology engineering
- Foundry efforts pursue this by aligning on core vocabulary and development principles
- While recognizing that minimizing downstream impact of misalignment requires governance around derived products used for specific applications

Bridge to Tradecraft

How do we build a worthwhile authoritative semantic layer?

- **Tradecraft** produces ontology content worth coordinating: scoped, justified, testable, and reusable
- By leveraging ontological insights in combination with semantic web technologies, as will see in later lectures

References

Derived Products from an Authoritative Semantics Layer

https://ncor-network.org/blog/derived-products-from-authoritative-semantics

Lossy, Never Divergent: Why Meaning Needs its Own Architecture

https://ncor-network.org/blog/lossy-never-divergent-semantic-architecture

The Round-Trip Test Every Data Platform Should Pass

https://ncor-network.org/blog/round-trip-test-semantic-accountability